

**Carnegie
Mellon
University**



VariAbility

ACCESSIBILITY IN COMPUTATION LABORATORY

Enhancing Learning Platforms for Individuals with ADHD

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VariAbility Lab HCI Research with Andrew Begel



About Me - Veronica

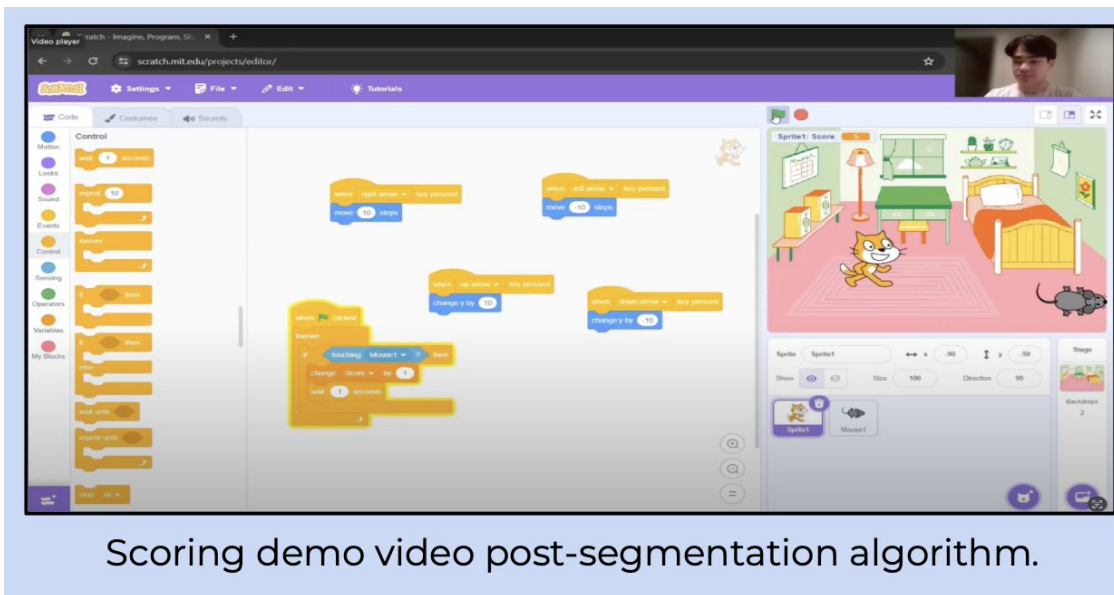
- Senior in Information Systems, HCI minor
- Incoming PhD student

Research Interests

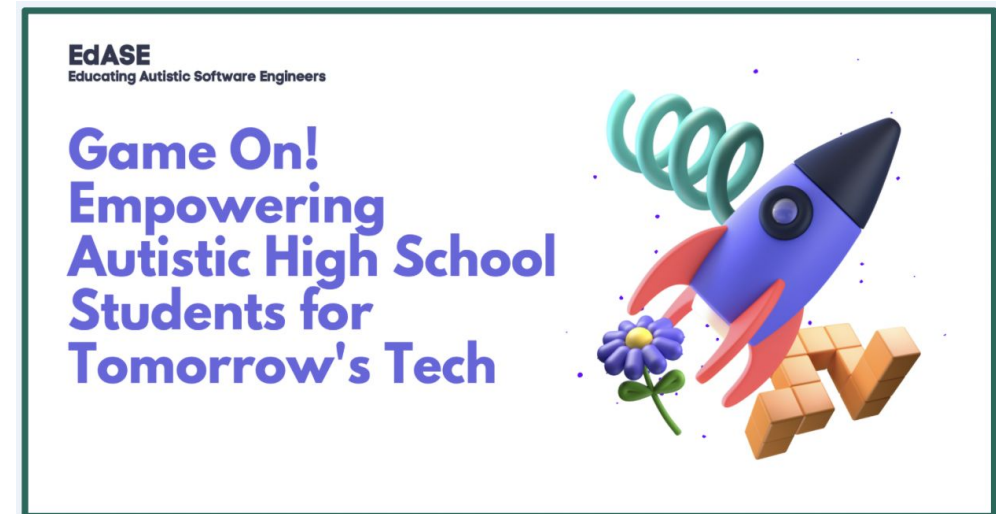
- Accessibility for minority groups- women & the neurodiverse
- Specifically applied to software engineering, programming education

NAPE Project

- How can a learning platform automatically customize instructional materials to accommodate different cognitive styles, such as ADHD, dyslexia, autism, etc.?



Scoring demo video post-segmentation algorithm.





Segmentation Intervention

- A pause of 4 seconds is added every time a “step” happens in the instructional video (instructor tells students to do an action)
- **Meant to reduce cognitive load for individuals with ADHD (based on Cognitive Load Theory)**
- Just submitted to ICER (yay)



Future Work & Interests

- Automated video segmentation
- Workplace accessibility (eg. slack for screen readers)
- Gender equality in software engineering
- Neurodiversity in general (chronic illness, adhd, autism, and more...)